MATH 200 - Elementary Linear Algebra

MATH 201 - Intermediate Linear Algebra

Section 7.1 Notes

Eigenvalues and Eigenvectors

Devotional Thought

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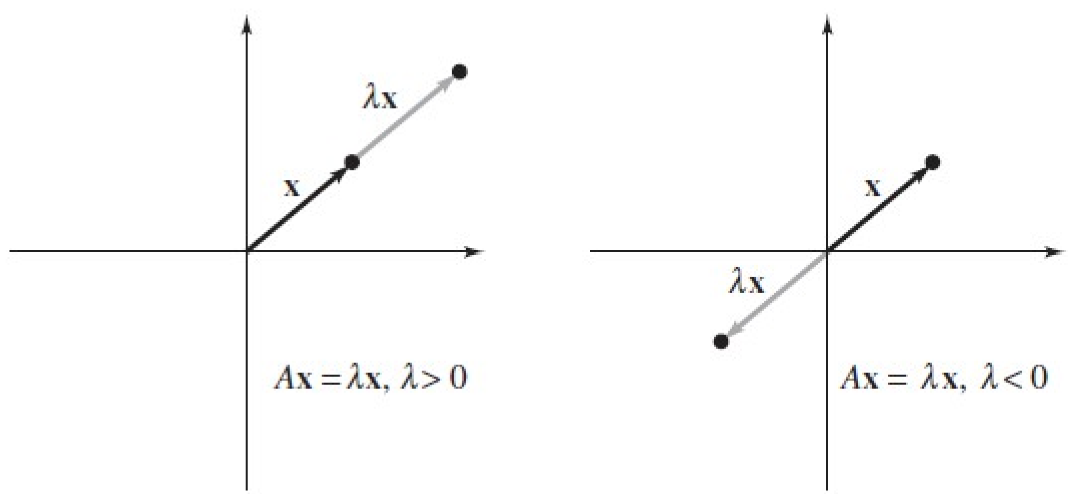
The Eigenvalue Problem

This section presents one of the most important problems in linear algebra, the **eigenvalue problem**. Its central question is "when A is an $n \times n$ matrix, do nonzero vectors $\mathbf{x}$ in R^n exist such that $A\mathbf{x}$ is a scalar multiple of $\mathbf{x}$?" The scalar, denoted by the Greek letter lambda (λ), is called an **eigenvalue** of the matrix A, and the nonzero vector $\mathbf{x}$ is called an **eigenvector** of A corresponding to λ. So, we have

$$A\mathbf{x} = \lambda\mathbf{x}.$$

Eigenvalues and eigenvectors have many important applications (such as solving systems of differential equations). For now, we will consider a geometric interpretation of the problem in R^2:

If λ is an eigenvalue of a matrix A and $\mathbf{x}$ is an eigenvector of A corresponding to λ, then multiplication of $\mathbf{x}$ by the matrix A produces a vector $\lambda\mathbf{x}$ that is parallel to $\mathbf{x}$, as shown below.



Definition Let A be an $n \times n$ matrix. The scalar λ is an **eigenvalue** of A when there is a *nonzero* vector $\mathbf{x}$ such that $A\mathbf{x} = \lambda\mathbf{x}$. The vector $\mathbf{x}$ is an *eigenvector* of A corresponding to λ.

Note that an eigen*vector* cannot be zero. Allowing $\mathbf{x}$ to be the zero vector would render the definition meaningless, since $A\mathbf{0} = \lambda\mathbf{0}$ is true for all real values of λ. An eigenvalue of $\lambda = 0$, however, is possible.

A matrix can also have more than one eigenvalue and every eigenvector has *one or more representative* eigenvectors corresponding to it.

Remark We will only be considering eigenvectors of real eigenvalues. Complex eigenvalues can occur in the context of real valued matrices, but handling such values requires more knowledge about complex vector spaces than we have time to go into in this course. (See chapter 8 of the online textbook.)

Example [Example 1] For the matrix

$$A = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}$$

verify that $\mathbf{x}_1 = (1, 0)$ is an eigenvector of A corresponding to the eigenvalue $\lambda_1 = 2$, and that $\mathbf{x}_2 = (0, 1)$ is an eigenvector of A corresponding to the eigenvalue $\lambda_2 = -1$.

Example [Example 2] For the matrix

$$A = \begin{bmatrix} 1 & -2 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

verify that

$$\mathbf{x}_1 = (-3, -1, 1) \quad \text{and} \quad \mathbf{x}_2 = (1, 0, 0)$$

are eigenvectors of A and find their corresponding eigenvalues.

Eigenspaces

Although both examples list only one eigenvector for each eigenvalue, each of the four eigenvalues given in the examples has infinitely many eigenvectors. For instance, in our first example, the vectors $(2, 0)$ and $(3, 0)$ are eigenvectors of A corresponding to the eigenvalue 2.

In fact, if A is an $n \times n$ matrix with an eigenvalue λ and a corresponding eigenvector $\mathbf{x}$, then every nonzero scalar multiple of $\mathbf{x}$ is also an eigenvector of A. To see this, let c be a nonzero scalar, which then produces

$$A(c\mathbf{x}) = c(A\mathbf{x}) = c(\lambda\mathbf{x}) = \lambda(c\mathbf{x}).$$

It is also true that if $\mathbf{x}_1$ and $\mathbf{x}_2$ are eigenvectors corresponding to the *same* eigenvalue λ, then their sum is also an eigenvector corresponding to λ, because

$$A(\mathbf{x}_1 + \mathbf{x}_2) = A\mathbf{x}_1 + A\mathbf{x}_2 = \lambda\mathbf{x}_1 + \lambda\mathbf{x}_2 = \lambda(\mathbf{x}_1 + \mathbf{x}_2).$$

In other words, the set of all eigenvectors of an eigenvalue λ, together with the zero vector, form a subspace of R^n.

Definition If A is an $n \times n$ matrix with an eigenvalue λ, then the set of all eigenvectors of λ, together with the zero vector

$$\{\mathbf{x} : \mathbf{x} \text{ is an eigenvector of } \lambda\} \cup \{\mathbf{0}\}$$

is a subspace of R^n. This subspace is the **eigenspace** of λ, often denoted E_λ.

That is, the eigenvectors of λ span the eigenspace of λ. This is turn means that by determining linearly independent eigenvectors we can find a basis for the eigenspace.

Determining the eigenvalues and corresponding eigenspaces of a matrix can involve algebraic manipulation. Occasionally, however, it is possible to find eigenvalues and eigenspaces by inspection.

Example [Example 3] Find the eigenvalues and corresponding eigenspaces.

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Note The geometric solution presented above is *not* typical of the general eigenvalue problem; we include it more for the sake of intuition.

Finding Eigenvalues and Eigenvectors

To find the eigenvalues and eigenvectors of an $n \times n$ matrix A, let I be the $n \times n$ identity matrix. Rewriting $A\mathbf{x} = \lambda\mathbf{x}$ as $\lambda I\mathbf{x} = A\mathbf{x}$ and rearranging gives $(\lambda I - A)\mathbf{x} = \mathbf{0}$. This homogeneous system of equations has nonzero solutions if and only if the coefficient matrix $(\lambda I - A)$ is *not* invertible—that is, if and only if its determinant is zero (FTELA).

Theorem 7.2 Let A be an $n \times n$ matrix.

1. An eigenvalue of A is a scalar λ such that $\det(\lambda I - A) = 0$.
2. The eigenvectors of A corresponding to λ are the nonzero solutions $\mathbf{x}$ of the vector equation $(\lambda I - A)\mathbf{x} = \mathbf{0}$.

The equation $\det(\lambda I - A) = 0$ is the **characteristic equation** of A. Moreover, when expanded to polynomial form, the polynomial

$$|\lambda I - A| = \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_2\lambda^2 + c_1\lambda + c_0$$

is the **characteristic polynomial** of A. So, the eigenvalues of an $n \times n$ matrix A correspond to the roots of the characteristic polynomial of A.

Remark The characteristic polynomial of A is of degree n, so A can have at most n distinct eigenvalues. The Fundamental Theorem of Algebra states that an nth-degree polynomial has precisely n roots. These n roots, however, include both repeated and complex roots. As a reminder, we will focus only on the real roots of characteristic polynomials—that is, real eigenvalues.

Example [Exercise 18] Find the eigenvalues and corresponding eigenvectors.

$$A = \begin{bmatrix} -2 & 4 \\ 1 & 1 \end{bmatrix}.$$

Process for Finding Eigenvalues and Eigenvectors

Let A be an $n \times n$ matrix.

1. Form the characteristic equation $|\lambda I - A| = 0$. It will be a polynomial equation of degree n in the variable λ.

2. Find the roots of the characteristic equation. These are the eigenvalues of A.

3. For each eigenvalue λ_i, find the eigenvectors corresponding to λ_i by solving the homogeneous system $(\lambda_i I - A)\mathbf{x} = \mathbf{0}$. This can require row reducing an $n \times n$ matrix. The reduced row-echelon form must have at least one row of zeros.

Notes

- Finding the eigenvalues of an $n \times n$ matrix involves the factorization of an nth-degree polynomial. (There is no foolproof method for achieving this.)

- Once you have found an eigenvalue, you can find the corresponding eigenvectors by any appropriate method for solving the under-determined system, such as Gauss-Jordan elimination. (There will be at least one free variable.)

- The homogeneous systems that arise when you are finding eigenvectors will always row reduce to a matrix having at least one row of zeros, because the systems must have nontrivial solutions. (The number of independent free variables is the dimension of the eigenspace.)

<u>Example</u> [Example 5] Find the eigenvalues and corresponding eigenvectors of A.

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

What is the dimension of the eigenspace for each eigenvalue?

An Important Aside Regarding Multiplicity

If an eigenvalue λ_i occurs as a *multiple root* (k times) of the characteristic polynomial, then λ_i has **multiplicity** k. This implies that $(\lambda - \lambda_i)^k$ is a factor of the characteristic polynomial and $(\lambda - \lambda_i)^{k+1}$ is not a factor of the characteristic polynomial. For instance, in the example above, the eigenvalue $\lambda =$ __ has a multiplicity of __.

Also note in the above, the dimension of the eigenspace of $\lambda =$ __ is __. In general, the *multiplicity of an eigenvalue is greater than or equal to the dimension of its eigenspace.*

<u>**Remark**</u> Multiplicity of an eigenvalue λ for a matrix A is often called **algebraic multiplicity** of λ, denoted $\mu_A(\lambda)$, while the dimension of its eigenspace is the **geometric multiplicity** of λ, denoted $\gamma_A(\lambda)$.

As can be observed from solving the characteristic equation, the sum of the algebraic multiplicities must be equal to the size of the matrix:

Given any square matrix A of size n with eigenvalues λ_1, λ_2, $\ldots$ λ_m (where $m \leq n$), it must be that

$$\sum_{i=1}^{m} \mu_A(\lambda_i) = n.$$

Example [Exercise 24] Find the eigenvalues of

$$A = \begin{bmatrix} 3 & 2 & -3 \\ -3 & -4 & 9 \\ -1 & -2 & 5 \end{bmatrix}$$

and find a basis for each of the corresponding eigenspaces.

Remark Finding eigenvalues and eigenvectors of matrices of order $n \geq 4$ can be tedious. Moreover, performing Gauss-Jordan elimination on a computer can introduce round off errors. Consequently, it can be more efficient (and accurate) to use numerical methods of approximating eigenvalues.

However, there is one type of matrix (of any order) whose eigenvalues are easily found.

Theorem 7.3 If A is an $n \times n$ triangular matrix, then its eigenvalues are the entries on its main diagonal.

Example [Example 7] Find the eigenvalues of each matrix.

a. $A = \begin{bmatrix} 2 & 0 & 0 \\ -1 & 1 & 0 \\ 5 & 3 & -3 \end{bmatrix}$

b. $A = \begin{bmatrix} -1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -4 & 0 \\ 0 & 0 & 0 & 0 & 3 \end{bmatrix}$

Eigenvalues and Eigenvectors of Linear Transformations

This section began with definitions of eigenvalues and eigenvectors in terms of matrices. Eigenvalues and eigenvectors can also be defined in terms of linear transformations. A number λ is an **eigenvalue** of a linear transformation T when there is a nonzero vector $\mathbf{x}$ such that $T(\mathbf{x}) = \lambda\mathbf{x}$. The vector $\mathbf{x}$ is an **eigenvector** of T corresponding to λ, and the set of all eigenvectors of λ (with the zero vector) is the **eigenspace** of λ.

Example [Example 8] Find the eigenvalues and a basis for each corresponding eigenspace of the linear transformation $T : R^3 \to R^3$ given by the standard matrix.

$$A = \begin{bmatrix} 1 & 3 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

The previous example illustrates an important connection between eigenvalues and eigenvectors, linear transformations, change of basis, and diagonalization:

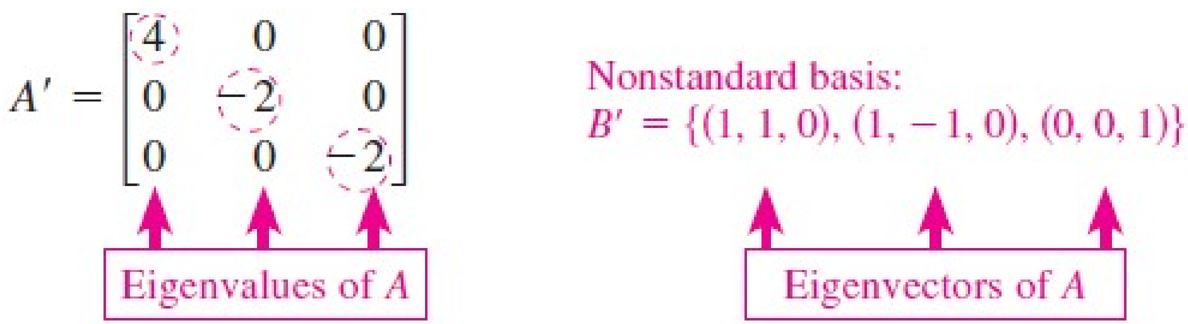
Consider $T : R^3 \to R^3$, whose matrix relative to the standard basis is

$$A = \begin{bmatrix} 1 & 3 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}. \qquad \text{Standard basis: } B = \{(1,0,0),(0,1,0),(0,0,1)\}$$

In Example 5 of Section 6.4 we found that the matrix of T relative to the basis $B' = \{(1,1,0),(1,-1,0),(0,0,1)\}$ is

$$A' = \begin{bmatrix} 4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}. \qquad \text{Nonstandard basis: } B' = \{(1,1,0),(1,-1,0),(0,0,1)\}$$

In the last example (Example 8), we saw that A has eigenvalues $\lambda = 4$ and $\lambda = -2$; these are precisely the entries in the diagonal matrix A', with -2 appearing twice correlating with the multiplicity of the eigenvalue being two. Moreover, the entries in the nonstandard basis are precisely the basis vectors for the eigenspaces.



This provides *the main idea of what eigenvectors actually are*: given a linear transformation with standard matrix A (relative to the standard basis) the eigenvectors form a natural basis B' (or coordinate system, if you will) for the range of T, one where the matrix A' for T relative to B' is diagonal.

Example [Exercise 48] Consider the linear transformation $T : R^3 \to R^3$ whose matrix A relative to the standard basis is given. Find (a) the eigenvalues of A, (b) a basis for each of the corresponding eigenspaces, and (c) the matrix A' for T relative to the basis B', where B' is made up of the basis vectors found in part (b).

$$A = \begin{bmatrix} 3 & 1 & 4 \\ 2 & 4 & 0 \\ 5 & 5 & 6 \end{bmatrix}$$